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DISPLAY PANEL AND DISPLAY APPARATUS

Abstract

A display panel includes a substrate, a light-emitting unit, and a touch component. The light-emitting unit is arranged on a side of the substrate, and has multiple first light-emitting units emitting light of a same color. The touch component is arranged on a side of the light-emitting unit away from the substrate, and has a touch trace. The touch trace has a first segment and a second segment. The first segment and the second segment are distributed along a periphery of a same first light-emitting unit. The first segment and the second segment extend in different directions. The first segment is provided with a first slit that extends through the first segment in its thickness direction. Multiple first segments on peripheries of multiple first light-emitting units extend in a same direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Chinese Patent Application No. 202410267207.9, filed on Mar. 8, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of display device and, in particular, to a display panel, a display apparatus, and a control method.

BACKGROUND

[0003] With the continuous development of technology, there is an increasing demand for display devices. In order to achieve a full-screen touch function, a touch metal layer is arranged over a light-emitting layer of a display device. Since the touch metal layer has an uneven shape and is located on one side of the light-emitting layer facing a light-exiting surface of the display device, it will affect the display effect of the display device.

SUMMARY

[0004] One aspect of the present disclosure provides a display panel. The display panel includes a substrate, a light-emitting unit, and a touch component. The light-emitting unit is arranged on a side of the substrate, and has multiple first light-emitting units emitting light of a same color. The touch component is arranged on a side of the light-emitting unit away from the substrate, and has a touch trace. The touch trace has a first segment and a second segment. The first segment and the second segment are distributed along a periphery of a same first light-emitting unit. The first segment and the second segment extend in different directions. The first segment is provided with a first slit that extends through the first segment in its thickness direction. Multiple first segments on peripheries of multiple first light-emitting units extend in a same direction.

[0005] Another aspect of the present disclosure provides a display apparatus. The display apparatus includes a display panel. The display panel includes a substrate, a light-emitting unit, and a touch component. The light-emitting unit is arranged on a side of the substrate, and has multiple first light-emitting units emitting light of a same color. The touch component is arranged on a side of the light-emitting unit away from the substrate, and has a touch trace. The touch trace has a first segment and a second segment. The first segment and the second segment are distributed along a periphery of a same first light-emitting unit. The first segment and the second segment extend in different directions. The first segment is provided with a first slit that extends through the first segment in its thickness direction. Multiple first segments on peripheries of multiple first light-emitting units extend in a same direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Upon reading the detailed description of the non-limiting embodiments provided below with reference to the accompanying drawings, other features, objects and advantages of the present disclosure will become more apparent and be encompassed with the scope of the present disclosure. Identical or similar reference numerals in the drawings may be used to denote identical or similar features.

[0007] FIG. 1 illustrates a schematic structural diagram of a display panel according to one embodiment of the present disclosure.

[0008] FIG. 2 illustrates a schematic diagram of a locally enlarged structure of FIG. 1 according to one embodiment of the present disclosure.

[0009] FIG. 3 illustrates a schematic diagram of a locally enlarged structure at P in FIG. 2.

[0010] FIG. 4 illustrates a cross-sectional view at A-A in FIG. 3.

[0011] FIG. 5 illustrates a schematic diagram of a locally enlarged structure of FIG. 3 according to one embodiment of the present disclosure.

[0012] FIG. 6 illustrates a schematic diagram of a locally enlarged structure of FIG. 3 according to another embodiment of the present disclosure.

[0013] FIG. 7 illustrates a schematic diagram of a locally enlarged structure at P in FIG. 2 according to another embodiment of the present disclosure.

[0014] FIG. 8 illustrates a schematic diagram of a locally enlarged structure of FIG. 7.

[0015] FIG. 9 illustrates a schematic diagram of a locally enlarged structure of FIG. 3 according to another embodiment of the present disclosure.

[0016] FIG. 10 illustrates a schematic diagram of a locally enlarged structure at Q in FIG. 2.

[0017] FIG. 11 illustrates a schematic diagram of a locally enlarged structure of FIG. 3 according to another embodiment of the present disclosure.

[0018] FIG. 12 illustrates a schematic diagram of a locally enlarged structure of FIG. 1 according to another embodiment of the present disclosure.

[0019] FIG. 13 illustrates a cross-sectional view at B-B in FIG. 12.

[0020] FIG. 14 illustrates a cross-sectional view at A-A in FIG. 3 according to another embodiment of the present disclosure.

[0021] FIG. 15 illustrates a schematic structural diagram of a display apparatus provided by the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0022] Features and exemplary embodiments of various aspects of the present disclosure are described in detail below. In the following detailed description, numerous specific details are outlined to provide a thorough understanding of the application. However, it will be apparent to those persons of ordinary skill in the art, that the present disclosure can be implemented without some of these specific details. The following description of embodiments is merely intended to provide a better understanding of the present disclosure by illustrating examples thereof. In the drawings and the following description, at least some well-known structures and techniques are not shown to avoid unnecessarily obscuring the present disclosure. The dimensions of some structures may be exaggerated for clarity. Furthermore, the features, structures, or characteristics described below may be combined in any suitable manner in one or more embodiments.

[0023] In the description of the present disclosure, it should be noted that, unless otherwise indicated, the terms such as “multiple” and “a plurality” mean two or more. Terms such as “upper”, “lower”, “left”, “right”, “inside”, “outside”, etc., indicating positional relationships or directions are for convenience in describing the present disclosure and simplifying the description, rather than indicating or implying that the device or component referred to have specific orientations, be constructed, or operated in specific directions, and thus cannot be understood as limiting the present application. Additionally, terms such as “first”, “second”, etc., are used solely for descriptive purposes and cannot be construed as indicating or implying relative importance.

[0024] The directional terms appearing in the following description are in accordance with the directions indicated in the figures and do not specifically limit the structure of embodiments of the present disclosure. Additionally, it should be noted in the description of the present disclosure that, unless otherwise explicitly specified and limited, terms such as “installation” and “connection” should be broadly construed. For example, they could mean a fixed connection, a detachable connection, or an integral connection. They can be directly connected or indirectly connected. The

specific meanings of the above terms in the present disclosure can be understood by those persons of ordinary skill in the art, depending on the specific circumstances.

[0025] A display panel includes a light-emitting unit and a touch trace located on the light-emitting unit. The touch trace includes an effective electrode and a dummy electrode. To avoid short-circuiting between the effective electrode and the dummy electrode, a slit is set between the effective electrode and the dummy electrode, causing them to be spaced apart and insulated from each other. When the display panel displays a monochrome image, diagonal display defects may occur. Further investigation reveals that due to the arrangement of the touch trace around the light-emitting unit, the shapes of the effective electrode and the dummy electrode may differ at different positions. This causes slits on the peripheries of light-emitting units of a same color to extend in different directions, leading to different light reflections at different positions under a large viewing angle. This results in a diagonal pattern in the monochrome image display.

[0026] To better convey the present disclosure, detailed descriptions of embodiments of the display panel and the display apparatus of the present disclosure are provided below with reference to FIGS. 1 to 15.

[0027] FIG. 1 illustrates a schematic diagram of a local structure of a display panel **10** provided by the present disclosure. FIG. 2 illustrates an enlarged schematic diagram of the local structure of FIG. 1. FIG. 3 is an enlarged schematic diagram of position P in FIG. 2. FIG. 4 is a partial cross-sectional view of FIG. 3.

[0028] As shown in FIGS. 1 to 4, the display panel **10** provided by embodiments of the present disclosure includes a substrate **100**, a light-emitting unit **200**, and a touch component **300**. The light-emitting unit **200** is arranged on a side of the substrate **100** and includes a plurality of first light-emitting units **210** that emit light of a same color. The touch component **300** is arranged on a side of the light-emitting unit **200** away from the substrate **100** and includes a touch trace **301**. The touch trace **301** includes a first segment **310** and a second segment **320** distributed around a same first light-emitting unit **210**. The first segment **310** and the second segment **320** extend in different directions. The first segment **310** is provided with a first slit **311** that extends through the first segment **310** along a width direction of the first segment **310**. A plurality of first segments **310** on peripheries of the plurality of first light-emitting units **210** extends in a same direction.

[0029] In the display panel **10** provided in embodiments of the present disclosure, the display panel **10** includes a substrate **100**, a light-emitting unit **200**, and a touch component **300**. The light-emitting unit **200** is located on a side of the substrate **100** to achieve the light-emitting display function of the display panel **10**. The touch component **300** is arranged on the side of the light-emitting unit **200** away from the substrate **100** to achieve the touch function of the display panel **10**. The touch component **300** includes a touch trace **301**. Light emitted from the first light-emitting unit **210** will exit through the touch trace **301**. Therefore, the touch trace **301** will affect the display effect of the first light-emitting unit **210**. The touch trace **301** includes the first segment **310** and the second segment **320** with different extension directions. The first segment **310** and the second segment **320** are distributed around a periphery of a first large light-emitting circle. A first slit **311** is provided on the first segment **310**. The first slit **311** will affect the large-angle light-emitting effect of the first light-emitting unit **210**. In embodiments of the present disclosure, an extension direction of a plurality of first segments **310** around a periphery of the plurality of first light-emitting units **210** is the same, i.e., an extension direction of a plurality of first slits **311** is the same, thereby making the light emission effects of the plurality of first light-emitting units **210** of a same color tend to be consistent, improving the display effect of the display panel **10**.

[0030] There are various ways to set the substrate **100**. The substrate **100** can include a supporting base and an array substrate set on the supporting base. The array substrate can include a pixel driving circuit.

[0031] There are various ways to set the supporting base, such as the supporting base can be a rigid supporting base, and a material of the supporting base includes a rigid material such as glass.

Alternatively, the supporting base can be a flexible supporting base, and a material of the supporting base includes a flexible material such as polyimide.

[0032] The array substrate may include a first conductive layer, a second conductive layer, and a third conductive layer that are set and stacked in sequence on a side of the supporting base. An insulating layer is provided between adjacent conductive layers. For example, the pixel driving circuit includes a transistor and a storage capacitor. The transistor includes a semiconductor, a gate electrode, a source electrode, and a drain electrode. The storage capacitor includes a first electrode plate and a second electrode plate. As an example, the gate electrode and the first electrode plate can be located on the first conductive layer. The second electrode plate can be located on the second conductive layer. The source electrode and the drain electrode can be located on the third conductive layer.

[0033] Optionally, the display panel **10** can be an organic light-emitting diode display panel. A pixel defining layer **500** is set on the substrate **100**. The pixel defining layer **500** includes a pixel defining part and a pixel opening **520** in the pixel defining part. The light-emitting unit **200** is located in the pixel opening **520**. Alternatively, the display panel **10** can also be a micro-light emitting diode display panel. A soldering pad is set on the substrate **100**. The light-emitting unit **200** includes a contact electrode. The light-emitting unit **200** is set on the substrate **100**. The contact electrode of the light-emitting unit **200** is electrically connected to the soldering pad. Alternatively, the display panel **10** can also be a liquid crystal display panel, where the light-emitting unit **200** includes a light source, a liquid crystal layer set on the light source, and a light filtering unit.

[0034] Optionally, an encapsulation layer **600** is provided on the side of the light-emitting unit **200** away from the substrate **100**. The encapsulation layer **600** is used to provide sealing protection to the light-emitting unit **200**. Optionally, the touch component **300** is disposed on a side of the encapsulation layer **600** away from the light-emitting unit **200** to improve the encapsulation effect of the touch component **300** on the encapsulation layer **600**.

[0035] Optionally, the light-emitting unit **200** can further include a second light-emitting unit **220** and a third light-emitting unit **230**. The light-emitting colors of the first light-emitting unit **210**, the second light-emitting unit **220**, and the third light-emitting unit **230** are different to achieve the color display of the display panel **10**. One of the first light-emitting unit **210**, the second light-emitting unit **220**, and the third light-emitting unit **230** is used to emit red light, another is used to emit green light, and the last one is used to emit blue light.

[0036] The touch trace **301** includes a first segment **310** and a second segment **320** distributed on the periphery of the same first light-emitting unit **210**. This means orthographic projections of the first segment **310** and the second segment **320** on the substrate **100** are located on the periphery of an orthographic projection of the first light-emitting unit **210** on the substrate **100**.

[0037] For example, the first segment **310** has a length direction and a width direction. An extension dimension of the first segment **310** in the length direction is greater than an extension dimension of the first segment **310** in the width direction. The first slit **311** traverses the first segment **310** in the width direction. Optionally, the first segment **310** can correspond to a portion of the touch trace **301** that is located on the periphery of the first light-emitting unit **210** and extending in a same direction. For example, the first segment **310** can correspond to a portion of the touch trace **301** that is located at the periphery of the first light-emitting unit **210** and extending in a same direction. Optionally, the second segment **320** can correspond to a portion of the touch trace **301** that is located at the periphery of the first light-emitting unit **210** and extending in a same direction. The extension directions of the second segment **320** and the first segment **310** are different.

[0038] As shown in FIG. 5, a first segment **310** may be provided on the periphery of a same first light-emitting unit **210**. The first segment **310** is located on one side of the first light-emitting unit **210**.

[0039] In some optional embodiments, as shown in FIG. 6, two or more first segments **310** with a same extension direction are provided on the periphery of the same first light-emitting unit **210**. In

some optional embodiments, first slits **311** are provided on more than two touch traces **301** with a same extension direction on the periphery of the same first light-emitting unit **210** to form more than two first segments **310**. This causes the light-emitting effect on the periphery of the same first light-emitting unit **210** to be more consistent.

[0040] Optionally, the touch trace **301** includes two first segments **310** oppositely arranged along the first direction X and two second segments **320** oppositely arranged along the second direction Y. Two first segments **310** and two second segments **320** are connected alternately end to end around the first light-emitting unit **210**, and the first direction X and the second direction Y intersect. This addresses the blocking of the touch trace **301** of the light-emitting unit **200** on the light-emitting effect of the display panel **10**. For example, two first segments **310** and two second segments **320** are alternately connected to form a hollow space. An orthographic projection of the first light-emitting unit **210** on the substrate **100** is located within an orthographic projection of the hollow space on the substrate **100** to improve the influence of the first segment **310** and the second segment **320** on the light-emitting effect of the first light-emitting unit **210**.

[0041] Optionally, the first direction X and the second direction Y are perpendicular. Optionally, the first direction X can be the width direction of the first segment **310**, and the second direction Y can be the length direction of the first segment **310**. Therefore, the first slit **311** can be set to traverse the first segment **310** along the first direction X. Optionally, the first direction X can be a length direction of the second segment **320**, and the second direction Y can be a width direction of the second segment **320**.

[0042] In some optional embodiments, as shown in FIG. 7, the light-emitting unit **200** further includes a plurality of second light-emitting units **220** with a same light-emitting color and a plurality of third light-emitting units **230** with a same light-emitting color. A first segment **310** is distributed on the periphery of the second light-emitting unit **220** and/or the third light-emitting unit **230**. That is to say, peripheries of the second light-emitting unit **220** and/or the third light-emitting unit **230** are also provided with first slits **311** extending in a same direction to ensure consistency in the light emission effects of the plurality of second light-emitting units **220** and/or the plurality of third light-emitting units **230**. This improves the light emission effect of the display panel **10**.

[0043] Alternatively, in other embodiments, as shown in FIG. 8, the touch trace **301** includes a third segment **330** distributed around a same second light-emitting unit **220** and/or a third light-emitting unit **230**. An extension direction of the third segment **330** is different from that of the first segments **310**. A second slit **331** is provided traversing along the width direction on the third segment **330**. In these optional embodiments, peripheries of the second light-emitting unit **220** and/or the third light-emitting unit **230** can correspondingly be provided with a second slit **331** that extends in a direction different than the first slit **311**. This ensures the pattern of the touch trace **301** to meet functional requirements.

[0044] Optionally, the extension direction of the third segment **330** can be the same as that of the second segment **320**. The difference between the third segment **330** and the second segment **320** is that the third segment **330** is provided with a second slit **331**, while the second segment **320** is not necessarily provided with a slit. That is, first slits **311** extending in a same direction are provided on first segments **310** that are on peripheries of the plurality of first light-emitting units **210**. Second segments **320** that are on the peripheries of the plurality of first light-emitting units **210** are not provided with slits. Directions of the slits on peripheries of the plurality of first light-emitting units **210** are consistent, thereby making the light-emitting effects of the plurality of first light-emitting units **210** more consistent.

[0045] In some optional embodiments, the first light-emitting unit **210** is a red light-emitting unit **200**. First slits **311** are provided on first segments **310** that are on peripheries of the plurality of red light-emitting units **200**. That is to say, the first slits **311** extending in a same direction are provided on the first segments **310** that are on peripheries of the plurality of red light-emitting units **200**.

Directions of the slits on the peripheries of the plurality of red light-emitting units **200** are consistent, thereby making the light emission effects of the plurality of red light-emitting units **200** more consistent.

[0046] Among a plurality of light-emitting units **200** with different emitting colors, the inconsistent light-emitting effect of the red light-emitting unit **200** is more perceptible to the human eye. The light-emitting effect of the red light-emitting unit **200** is more likely to affect the overall light-emitting effect of the display panel **10**. In one exemplary embodiment, by ensuring consistency in the direction of slits on the peripheries of the plurality of red light-emitting units **200**, the light-emitting effects of the plurality of red light-emitting units **200** are more consistent, which can better improve the display effect of the display panel **10**.

[0047] In some other optional embodiments, as shown in FIG. **9**, the light-emitting unit **200** further includes a red light-emitting unit **200** that emits light in a different color from the first light-emitting unit **210**. The touch trace **301** includes a ring segment located on the periphery of the red light-emitting unit **200** that is in a closed ring shape.

[0048] In these optional embodiments, the light-emitting color of the first light-emitting unit **210** is not red. The light-emitting unit **200** further includes a red light-emitting unit **200**. The touch trace **301** located around the red light-emitting unit **200** forms a closed ring shape, that is, there are no slits on the touch trace **301** around the red light-emitting unit **200**, making the light-emitting effects of the red light-emitting unit **200** at different positions more consistent. This can further improve the display effect of the display panel **10**.

[0049] In one exemplary embodiment, as shown in FIGS. **2** to **5**, if there are slits around the red light-emitting unit **200**, the slits around the red light-emitting unit **200** are all first slits **311**. The slits on the peripheries of the red light-emitting unit **200** extend in a same direction; Alternatively, as shown in FIG. **9**, no slits are provided on the periphery of the red light-emitting unit **200**, thereby making the light-emitting effects of red light-emitting units **200** at different positions more consistent and improving the display effect of the display panel **10**.

[0050] In some optional embodiments, quantities of first slits **311** on a same side of the periphery of two or more first light-emitting units **210** is equal. This makes the light-emitting effects on peripheries of different first light-emitting units **210** more consistent, which can better improve the display effect of the display panel **10**.

[0051] Optionally, the quantities of first slits **311** on the same side of two or more first light-emitting units **210** are equal, making the light-emitting effects on the periphery of different first light-emitting units **210** more consistent.

[0052] Optionally, as shown in FIGS. **2** and **10**, the touch component **300** includes a plurality of touch electrodes **340**. A touch electrode **340** includes a touch trace **301**. The touch electrode **340** can be a self-capacitive touch electrode or a mutual-capacitive touch electrode. Optionally, the touch trace **301** forms a grid pattern, and by setting separation slits **350** at different positions along the touch trace **301**, the touch trace **301** is divided into a plurality of touch electrodes **340**.

[0053] In some optional embodiments, as further illustrated in FIGS. **2** and **10**, the touch trace **301** includes a plurality of touch electrodes **340**. At least two adjacent touch electrodes **340** are spaced apart to form a separation slit **350**. At least one first slit **311** is reused as the separation slit **350**.

[0054] In these optional embodiments, a separation slit **350** needs to be set on the touch trace **301** to insulate a plurality of touch electrodes **340** from each other. At least one first slit **311** is reused as the separation slit **350**, i.e., the first slit **311** can be used to divide the touch trace **301** into a plurality of touch electrodes **340**. Reusing the first slit **311** and the separation slit **350** can reduce the total quantity of slits on the touch trace **301**, improving the display effect of the display panel **10**.

[0055] In some optional embodiments, as further illustrated in FIG. **10**, the touch electrode **340** includes an effective electrode **341** and a dummy electrode **342**. The effective electrode **341** is used to realize a touch function of the display panel **10**. The dummy electrode **342** can be used to fill

vacancies between effective electrodes **341** or inside an effective electrode **341** to improve the display effect of the display panel **10** in the screen-off state.

[0056] In some optional embodiments, the separation slit **350** can be used to separate two effective electrodes **341**, or the separation slit **350** can be used to separate the effective electrode **341** from the dummy electrode **342**. For example, in some embodiments, the dummy electrode **342** includes at least one first segment **310**, where a separation slit **311** can be set on the dummy electrode **342** to improve the effect of the first slit **311** on the touch function of the effective electrode **341**.

[0057] In some optional embodiments, the effective electrode **341** includes a touch sensing electrode **341a** and/or a touch driving electrode **341b**. This allows the effective electrode **341** to realize the touch function through the touch sensing electrode **341a** and/or the touch driving electrode **341b**. Optionally, the touch electrode **340** is a mutual-capacitive touch electrode **340**. The effective electrode **341** can include a touch sensing electrode **341a** and a touch driving electrode **341b**. In other embodiments, the touch electrode **340** can also be a self-capacitive touch electrode **340**. The effective electrode **341** is a touch electrode **340** block arranged separately and connected to a touch processor.

[0058] In some optional embodiments, as shown in FIG. **11**, at least one touch electrode **340** includes a through-hole **343** that traverses the touch electrode **340**. By setting the through-hole **343** on the touch electrode **340**, a distribution area of the touch electrode **340** can be reduced, thereby addressing the parasitic capacitance between the touch electrode **340** and a conductive layer over the substrate **100**, and enhancing the touch sensitivity.

[0059] Optionally, a plurality of through-holes **343** are spaced apart and distributed on the touch electrode **340**. This further reduces the distribution area of the touch electrode **340**, better addressing the issue of parasitic capacitance and improving touch sensitivity.

[0060] In some optional embodiments, as shown in FIGS. **12** and **13**, the display panel **10** includes a first touch layer **410** and a second touch layer **420**. A plurality of touch electrodes **340** is provided in the first touch layer **410**. The second touch layer **420** is provided with a bridging part **421**. At least two adjacent and spaced apart touch electrodes **340** are connected to each other through the bridging part **421**. For example, two adjacent touch sensing electrodes **341a** can be connected to each other through the bridging part **421**. Alternatively, two adjacent touch driving electrodes **341b** can be connected to each other through the bridging part **421**. Optionally, a first insulating layer **710** is provided between the first touch layer **410** and the second touch layer **420**. A via hole **711** is provided in the first insulating layer **710**. The touch electrode **340** can connect a via hole **711** of the bridging part **421** by the via hole **711** of the touch electrode **340**. The positions of the via hole **711** and the bridging part **421** are illustrated with dotted lines in FIG. **12**.

[0061] Optionally, the touch electrode **340** includes a touch area **344** and a bridging area **345**. An orthographic projection of the bridging area **345** on the substrate **100** overlaps with an orthographic projection of the bridging part **421** on the substrate **100**. An orthographic projection of the touch area **344** on the substrate **100** is located outside the orthographic projection of the bridging part **421** on the substrate **100**. The first segment **310** is located in the touch area **344**.

[0062] The touch trace **301**, on the touch electrode **340** and overlapping with the bridging part **421**, is usually specially designed. That is, the touch trace **301** inside the bridging area **345** is usually specially arranged, and the bridging area **345** occupies a relatively small proportion of the total area of the touch electrode **340**. Therefore, the first segment **310** is disposed in the touch area **344**. This is attributed to a larger area of the touch area **344**, ensuring the display effect by arranging the first segment **310** in the touch area **344**. Avoiding placing the first segment **310** in the bridging area **345** can simplify the structure of the touch trace **301**. In some optional embodiments, as shown in FIG. **14**, at least a portion of the surface of the touch trace **301** away from the light-emitting unit **200** is provided with a protrusion structure **301a**.

[0063] In these optional embodiments, by setting the protrusion structure **301a** on the surface of the touch trace **301**, light can undergo diffuse reflection on the outer surface of the touch trace **301**.

This improves the display effect of the display panel **10** by reducing the reflection of ambient light on the touch trace **301**.

[0064] Optionally, plasma etching or similar treatments can be applied to the touch trace **301** to form a pit structure and a protrusion structure **301a** on the surface of the touch trace **301** away from the light-emitting unit **200**. Alternatively, a second insulating layer **720** can be set on one side of the touch trace **301** facing the substrate **100**. One surface of the second insulating layer **720** away from the substrate **100** can be patterned to form a groove **721** and a convex part **722**. Then, a conductive material is deposited on one surface of the insulating layer with the groove **721** to form the touch trace **301**. The conductive material deposited in the groove **721** forms the pit structure on one surface of the touch trace **301**, while the conductive material deposited on the convex part **722** outside the groove **721** forms the protrusion structure **301a** on one surface of the touch trace **301**.

[0065] Optional, when the first touch layer **410** is located on one side of the second touch layer **420** away from the substrate **100**, the second insulating layer **720** and the first insulating layer **710** that is located between the first touch layer **410** and the second touch layer **420** can be reused with each other. In other embodiments, when the first touch layer **410** is located on one side of the second touch layer **420** facing the substrate **100**, the second insulating layer **720** can also serve as another insulating layer. The second insulating layer **720** and the first insulating layer **710** are different insulating layers.

[0066] Optionally, an entire touch trace **301** can be bent and protruded along a thickness direction Z to form the protrusion structure **301a**. Alternatively, the protrusion structure **301a** can be formed on the surface of the touch trace **301** away from the substrate **100**, while the surface of the touch trace **301** facing the substrate **100** is flat. As long as the surface of the touch trace **301** away from the substrate **100** has the protrusion structure **301a**, the protrusion structure **301a** is capable of reflecting light.

[0067] In some optional embodiments, the light-emitting unit **200** includes a red light-emitting unit **200**, and at least a portion of one surface of the touch trace **301** located on the periphery of the red light-emitting unit **200** has the protrusion structure **301a**.

[0068] In these optional embodiments, the protrusion structure **301a** is arranged on at least a portion of the touch trace **301** on the periphery of the red light-emitting unit **200**. Thus, light can undergo diffuse reflection on the periphery of the red light-emitting unit **200** to address the issue of reflected light on the periphery of the red light-emitting unit **200**. This makes the light-emitting effect of the plurality of the red light-emitting units **200** more consistent, thereby improving the display effect of the display panel **10**.

[0069] There are various locations for arranging the protrusion structure **301a**. The protrusion structure **301a** can be located on at least one of the first segment **310**, the second segment **320**, and the third segment **330**.

[0070] In some optional embodiments, the protrusion structure **301a** is set on the first segment **310**, and the protrusion structure **301a** is located on both sides of the first slit **311** in an extension direction of the first segment **310**.

[0071] In these optional embodiments, due to the presence of the first slit **311** on the first segment **310**, the location of the first slit **311** will not produce light reflection, while light reflection will occur on the first segment **310** that is located on both sides of the first slit **311**. This results in significant differences in display between the first slit **311** and first segments **310** which are located on both sides of the first slit **311**. By setting protrusion structures **301a** on both sides of the first slit **311** in the extension direction of the first segment **310**, the display difference between the first slit **311** and its periphery can be addressed, thereby better improving the display effect of the display panel **10**.

[0072] Optionally, the protrusion structures **301a** on both sides of the first slit **311** in the extension direction of the first segment **310** are symmetrically arranged with respect to the first slit **311**. Thus, this makes the display effect on both sides of the first slit **311** more uniform, thereby better

improving the display effect of the display panel **10**.

[0073] Optionally, quantities of the protrusion structures **301a** on both sides of the first slit **311** in the extending direction of the first segment **310** are consistent. This makes the display effect on both sides of the first slit **311** more uniform, thereby better improving the display effect of the display panel **10**.

[0074] Optionally, as described above, when an extension length of the first segment **310** in the first direction X is greater than its extension length in the second direction Y, the extension direction of the first segment **310** is the first direction X. The first slit **311** can extend along the second direction Y. When the extension length of the first segment **310** in the second direction Y is greater than its extension length in the first direction X, the extension direction of the first segment **310** is the second direction Y. The first slit **311** can extend along the first direction X.

[0075] In some optional embodiments, multiple protrusion structures **301a** are spaced apart along the extension direction of the touch trace **301**. For example, the touch trace **301** can be in a wavy structure, or one surface of the touch trace **301** away from the substrate **100** has a wavy structure to form the multiple protrusion structures **301a** spaced apart along their extension direction.

[0076] In these optional embodiments, by setting multiple protrusion structures **301a** and having the multiple protrusion structures **301a** spaced apart along the extension direction of the touch trace **301**, the touch trace **301** can diffusely reflect light at different positions. This improves the display effect of the display panel **10**.

[0077] In some optional embodiments, the touch trace **301** encloses a hollow area **302b**. The orthographic projection of the light-emitting unit **200** on the substrate **100** is located in an orthographic projection of the hollow area **302b** on the substrate **100**. The first segment **310** is a portion of the touch trace **301** that surrounds the hollow area **302b**. For example, two first segments **310**, and two second segments **320** can be alternately distributed and connected to each other to form the hollow area **302b**. Alternatively, two segments **310** and two third segments **330** can be alternately distributed and connected to each other to form the hollow area **302b**.

[0078] In these optional embodiments, the touch trace **301** encloses the hollow area **302b**. Light emitted by the light-emitting unit **200** can exit through the hollow area **302b** to improve the light output of the touch trace **301** on the light-emitting unit **200**.

[0079] In some optional embodiments, one first slit **311** is set on a same first segment **310**, or a plurality of first slits **311** are set on a same first segment **310**. The plurality of first slits **311** is spaced apart along the extension direction of the first segment **310**.

[0080] In embodiments of the present disclosure, one first slit **311** can be set on the first segment **310**, or a plurality of first slits **311** can be set on the first segment **310**, to enrich the arrangement of the first slits **311**. By reasonably setting the quantity and the position of the first slits **311**, the purpose of separating the touch trace **301** into a plurality of mutually insulated touch electrodes **340** can be achieved, and the display effect of the display panel **10** can also be improved.

[0081] As shown in FIG. **15**, embodiments of the present disclosure further provide a display apparatus. The apparatus includes the display panel **10** according to any one of the embodiments described above. The display apparatus has the beneficial effects of any one of the embodiments described above, which are not repeated here.

[0082] The display apparatus in embodiments of the present disclosure includes but is not limited to apparatuses with display functions such as mobile phones, personal digital assistant (PDA), tablets, e-books, televisions, access control systems, smart landlines, and consoles.

[0083] Although the present disclosure has been described with reference to preferred embodiments, various modifications may be made and equivalents may be substituted for components thereof without departing from the scope of the present disclosure. In particular, as long as there is no fundamental conflict, the technical features mentioned in various embodiments can be combined in any way. The present disclosure is not limited to specific embodiments disclosed herein but includes all technical solutions falling within the scope of the claims.

Claims

1. A display panel, comprising a substrate; a light-emitting unit, arranged on a side of the substrate, wherein the light-emitting unit comprises a plurality of first light-emitting units that emits light of a same color; and a touch component, arranged on a side of the light-emitting unit away from the substrate, wherein the touch component comprises a touch trace; the touch trace comprises a first segment and a second segment; the first segment and the second segment are distributed along a periphery of a same first light-emitting unit of the plurality of first light-emitting units; the first segment and the second segment extend in different directions; and the first segment is provided with a first slit that extends through the first segment in a thickness direction of the first segment; and a plurality of first segments on peripheries of the plurality of first light-emitting units extends in a same direction.
2. The display panel according to claim 1, wherein on a periphery of a same first light-emitting unit, more than two first segments are arranged extending in a same direction.
3. The display panel according to claim 2, wherein the touch trace further comprises two first segments arranged oppositely to each other along a first direction and two second segments arranged oppositely to each other along a second direction; the two first segments and the two second segments are alternately connected end-to-end around the first light-emitting unit; and the first direction intersects with the second direction.
4. The display panel according to claim 1, wherein the light-emitting unit further comprises a plurality of second light-emitting units that emits light of a same color and a plurality of third light-emitting units that emits light of a same color; and the first segment is distributed on a periphery of a second light-emitting unit of the plurality of second light-emitting units and/or a third light-emitting unit of the plurality of third light-emitting units; or the touch trace comprises a third segment distributed on a periphery of a same second light-emitting unit and/or a same third light-emitting unit; an extension direction of the third segment is different from an extension direction of the first segment; and the third segment is provided with a second slit that extends through the third segment in a thickness direction of the third segment.
5. The display panel according to claim 1, wherein a first light-emitting unit of the plurality of first light-emitting units is a red light-emitting unit; and the first segment on a periphery of a plurality of red light-emitting units are provided with the first slit; or the light-emitting unit further comprises a red light-emitting unit emitting light in a different color than the first light-emitting unit; the touch trace comprises a ring segment located on a periphery of a red light-emitting unit of the plurality of red light-emitting units; and the ring segment is in a closed ring shape.
6. The display panel according to claim 1, wherein quantities of first slits on the periphery of the plurality of first light-emitting units are identical.
7. The display panel according to claim 1, wherein the touch trace comprises a plurality of touch electrodes; at least two adjacent touch electrodes are spaced apart to form a separation slit; and at least one first slit is reused as the separation slit.
8. The display panel according to claim 7, wherein the plurality of touch electrodes comprises an effective electrode and a dummy electrode; and the dummy electrode comprises at least one first segment.
9. The display panel according to claim 8, wherein the effective electrode comprises a touch sensing electrode and/or a touch driving electrode.
10. The display panel according to claim 7, wherein at least one touch electrode comprises a through-hole that traverses the touch electrode; and a plurality of through-holes are spaced apart on the touch electrode.
11. The display panel according to claim 7, further comprising a first touch layer and a second touch layer, wherein the plurality of touch electrodes is located on the first touch layer; the second

touch layer is provided with a bridging part; and at least two adjacent touch electrodes are electrically connected to each other through the bridging part; and a touch electrode of the plurality of touch electrodes comprises a touch area and a bridging area; an orthographic projection of the bridging area on the substrate overlaps with an orthographic projection of the bridging part on the substrate; an orthographic projection of the touch area on the substrate is located outside the orthographic projection of the bridging part on the substrate; and the first segment is located in the touch area.

12. The display panel according to claim 1, wherein at least a portion of a surface of the touch trace away from the light-emitting unit is provided with a protrusion structure.

13. The display panel according to claim 12, wherein the light-emitting unit comprises a red light-emitting unit; and at least a portion of a surface of the touch trace located on a periphery of the red light-emitting unit includes the protrusion structure.

14. The display panel according to claim 12, wherein the protrusion structure is provided on the first segment; and the protrusion structure is located on both sides of the first slit in an extension direction of the first segment.

15. The display panel according to claim 12, further comprising multiple protrusion structures, wherein the multiple protrusion structures are spaced apart along an extension direction of the touch trace.

16. The display panel according to claim 1, wherein the touch trace encloses to form a hollow area; an orthographic projection of the light-emitting unit on the substrate is located in an orthographic projection of the hollow area on the substrate; and the first segment is a portion of the touch trace that encloses to form the hollow area.

17. The display panel according to claim 1, wherein one first slit is provided on a same first segment; or multiple first slits are provided on the first segment and are spaced apart in an extension direction of the first segment.

18. A display apparatus, comprising: a display panel, comprising a substrate; a light-emitting unit, arranged on a side of the substrate, wherein the light-emitting unit comprises a plurality of first light-emitting units that emits light of a same color; and a touch component, arranged on a side of the light-emitting unit away from the substrate, wherein the touch component comprises a touch trace; the touch trace comprises a first segment and a second segment; the first segment and the second segment are distributed along a periphery of a same first light-emitting unit of the plurality of first light-emitting units; the first segment and the second segment extend in different directions; and the first segment is provided with a first slit that extends through the first segment in a thickness direction of the first segment; and a plurality of first segments on peripheries of the plurality of first light-emitting units extends in a same direction.
